

HRISHIRAJ MANDAL

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EDUCATION

National University of Singapore (NUS)

August 2023 - May 2027

Bachelor's, Computer Science

GPA: 4.4

- Awards: Winner Ripple X NUS Fintech Summit Hackathon, Nels Radley Freits Smartkarma Fellowship
- Languages: Python, Java, JavaScript, Typescript, Solidity, C, C++, Scheme, OCaml
- Core Modules: Software Engineering, Algorithms and Data Structures, Information Security, Computer Networking, Database Systems, Programming Language Development, AI/ML, Operating Systems
- Semester projects: Microcontroller development using C, Arduino and Fusion360

PROFESSIONAL EXPERIENCE

National University of Singapore (NUS)

Singapore

Research and Teaching Assistant

May 2025 - Present

- Research Assistant, ACSL-SL conversion for Frama-C (Dec – Jan 2026): Designed and implemented a specification compiler in OCaml translating Separation Logic to ACSL for Frama-C, enabling automated formal verification of low-level C programs
- Teaching Assistant, Introduction to Information Security (Aug 2025 - Present): Conducted weekly tutorials for 40 students, explaining cryptography, access control, and vulnerability analysis; supported student consultations
- Teaching Assistant, Independent Software Development Project (May - June 2025): Mentored 13 student teams through end-to-end software projects (requirements, design, implementation, evaluation), leading 1 team to top 5% of the cohort

The Agency for Science, Technology and Research

Singapore

AI Research Intern

January 2025 - July 2025

- ARIA awardee - Contributed research on 'Demand Forecasting Frameworks for Fast Moving Consumer Goods (FMCG)'
- Performed systematic literature review on classical and machine-learning-based forecasting methods; synthesized findings to inform model selection and experimental design for publication-quality work
- Designed and evaluated forecasting pipelines using machine-learning time-series models within the Nixtla ecosystem (StatsForecast / MLForecast / NeuralForecast), improving robustness and interpretability of predictions
- Lead development of internal Explainable AI (XAI) package for demand-forecasting models, enabling feature-level and temporal attribution; collaborated in an agile, Git-based research workflow with cross-functional stakeholders
- Link to testimonial: https://www.hrishiraj.site/assets/astar_testimonial%20and%20cert-BZDhLwBy.pdf

Smartkarma

Singapore

Data Analyst Intern

February 2023 - July 2023

- Streamlined financial data collection pipelines using the Bloomberg API, reducing manual data extraction for analysis
- Built and optimized data dashboards by querying PostgreSQL and visualizing insights in Excel, supporting data-driven reporting
- Analyzed user behavior using Google Analytics 4, providing actionable recommendations for performance optimization
- Implemented marketing workflow automation using Zapier, improving outreach efficiency
- Testimonial of positive performance evaluation: https://www.hrishiraj.site/assets/SK-general-2xJmd_RW.pdf

PROJECTS & OUTSIDE EXPERIENCE

Oma

Singapore

Backend, Quality Assurance, DevOps lead

May 2024 - August 2024

- Led backend development for a native AI-powered dining recommendation application
- Implemented a recommendation system in PyTorch, supporting personalized dining suggestions based on user preferences
- Designed a type-safe full-stack architecture using Next.js, tRPC, and Turborepo, enabling shared types, modular services, and efficient monorepo development designing scalable services using established software architecture and design patterns
- Established automated testing and CI/CD pipelines using Vitest, enforcing code quality
- Deployed and operated the native application and ML inference services on Microsoft Azure, managing cloud resources

[Link to Project](#)

BARE

Singapore

Team Lead

November 2023 - December 2023

- Led a winning team submission at the Ripple Track, NUS FinTech Summit 2024 Hackathon, delivering a functional dApp
- Designed and implemented a decentralized rental agreement platform connecting landlords and tenants via Ethereum smart contracts, ensuring transparency and trustless execution; automating contract lifecycle using the Python Brownie framework
- Built the full-stack dApp interface using React.js, integrating on-chain logic with user-facing workflows

[Link to Project](#)

SKILLS

Skills: Python, Java, JavaScript, TypeScript, Solidity, C#, Pytorch, Next.js, Prisma, MongoDB, Postgres, Excel/Numbers/Sheets, Zapier, SQL, Playwright, Microsoft Azure, Blockchain, React.js, Data Structures & Algorithms, Computer Networking, Data Analysis, Machine Learning, Quality Assurance (QA), Development Operations (DevOps), Market Research

Languages: English (Native), Bengali (Expert), Mandarin (Conversational)